

Problem-Solution Fit

Date	15 July 2026
Team ID	SWTID-2026-2749
Project Name	OptiCrop: Smart Agricultural Production Optimization engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Farmers • Agricultural Officers • Agriculture Students • Smart Farming Organizations	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited technical knowledge • Internet availability • Budget constraints • Lack of scientific support	5. AVAILABLE SOLUTIONS PROS & CONS [AS] <i>Traditional farming advice</i> Pros • Experienced guidance Cons • Less accurate • Time consuming • No AI prediction
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Wrong crop selection • Low agricultural productivity • Poor soil analysis • Unpredictable weather • Financial losses	9. PROBLEM ROOT / CAUSE [RC] • Lack of data-driven decisions • Soil nutrient imbalance • Weather uncertainty • No AI assistance	7. BEHAVIOR + ITS INTENSITY [BE] • Consult local experts • Search on internet • Use previous farming experience

3. TRIGGERS TO ACT [TR]

- Soil nutrient imbalance
- Changing climate
- Need for higher yield
- Better farming decisions

4. EMOTIONS BEFORE [EM]

- Confused
- Worried
- Uncertain

AFTER

- Confident
- Satisfied
- Motivated

10. YOUR SOLUTION [SL]

- OptiCrop uses Machine Learning
- to analyze soil nutrients and
- environmental conditions to
- recommend the best crop along
- with season, soil type,
- temperature range and water
- requirement.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Agriculture Websites
- Government Portals
- Mobile Apps
- YouTube

OFFLINE

- Agricultural Officers
- Local Markets
- Farmer Meetings